



Pipeline leak detection using hydraulic transients and domain-guided machine learning

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ABSTRACT

In this paper, we present a novel domain-guided framework for training machine learning (ML) models, particularly neural networks (NN), to detect multiple leaks in pipelines using hydraulic transient datasets. Central to our approach is the utilization of a ‘leak function’ as the target variable in the ML model, where leak properties exhibit a pulse-like signature. This framework allows ML models to be trained without the typical constraints on the number of leaks and avoids the necessity of retraining separate models for different scenarios. To enhance model robustness against operational uncertainties, we employ iterative refinement techniques. Starting with a baseline NN model, each subsequent refinement incorporates factors such as valve closure time, pre-transient flow properties, and the capacity to handle an increased number of leaks. We validate the accuracy of our framework through numerical simulations and experimental data from a laboratory-scale Viscoelastic pipe system. Furthermore, we demonstrate the framework’s applicability to pipeline networks in two configurations: a series connection of two pipes and a tree-type network comprising three pipes. Our results indicate satisfactory detection accuracy and system adaptability, showcasing the potential of our framework for practical ML applications in leak detection.

1. Introduction

Pressurized pipelines are vital for water distribution to cities and industries, yet they are vulnerable to leaks and other faults (e.g., partial blockages, and wall deterioration). A leaking pipeline does not merely represent a physical fault in the system; it signifies a contributor to non-revenue water (NRW) losses, waste of energy, and a burden on financial resources. Globally, NRW accounts for approximately 346 million cubic meters per day [1]. A significant portion of this volume is attributed to leaks, reinforcing the urgency for effective solutions. Thus, the development of robust leak (fault) detection methods has emerged as an active and pressing area of research. The impetus for such advancements is twofold: on one hand, it addresses the immediate challenges faced by water agencies tasked with the management and maintenance of these pipelines; on the other, it presents an opportunity for broader economic benefits.

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Current commercial methodologies for pipeline leak (fault) detection, including noise loggers [2], listening rods [3], ground penetrating radar [4], the SAHARA system [5], and smart-ball [6], often face challenges with large-scale pipelines. These methods tend to be disruptive and intrusive, necessitating significant operational and financial expenditure for water supply agencies. From a management viewpoint, these methods are usually carried out by the supplier's technicians, which can lead to waiting times and delays. Moreover, the majority of these techniques are passive; they depend on the manifestations of existing leaks, such as noise or flow disturbances, to signal their presence [7]. This approach inherently limits their effectiveness, as they react to symptoms of leaks rather than actively detecting them. Consequently, despite the availability of these technologies, the persistent issue of NRW losses indicates a gap in the ability of current methods to detect and mitigate leaks efficiently.

Transient-based leak (fault) detection methods [8] are receiving immense attention as a viable strategy for identifying leaks and other faults. Leveraging mechanically induced hydraulic transient waves, which travel at the speed of sound – approximately 1000 m/s in metallic pipes and 400 m/s in non-metallic pipes [9] – this approach provides a non-disruptive and non-intrusive alternative suitable for large-scale pipelines. As these waves travel along the pipeline, they reflect off physical discontinuities, including leaks, sending back signals that carry distinctive information about these features. Precisely, leaks or junctions for side-branches generate a negative pressure wave, while a partially closed valve introduces a positive pressure wave. The characteristic pattern of an extended partial blockage is a positive wave followed by a negative one, forming a 'positive bell' profile. Conversely, partial wall deterioration is indicated by a negative wave succeeded by a positive one, creating a 'negative bell' pattern. The time-of-arrival and magnitude of these reflected pressure waves are associated respectively with the location and size of discontinuity. Therefore, by analyzing the measured transient responses, it becomes possible to decode these signals, pinpointing the presence, location, and size of leaks and other faults with precision. In addition, the short duration and simplicity of the transient tests give the water system technicians complete autonomy and guarantee minimum costs. However, the task is not straightforward as the measured signals are inevitably contaminated by environmental noise [10–14].

To deal with noise, the literature has traditionally focused on three primary approaches: model-based, data-driven (ML), and hybrid approaches. Model-based methods rely on physics-driven and inversion-based techniques [15], including inverse transient analysis [10,16–19], time-domain reflectometry [20], spectral-based techniques [21–28], and time-reversal methods [13,29,30]. Most of these methods have undergone rigorous testing predominantly in controlled laboratory environments with few attempts in the real pipe systems [31–35].

While the research community continues to extend and enhance this modeling and inversion-based approaches, there is a parallel and growing interest in utilizing data-driven machine learning (ML) for the same task. For instance, in 2019, Zhou et al. [36] utilized resonant frequency responses to train a Convolutional Neural Network (CNN) using a novel fusion-enhanced stochastic optimization algorithm combined with a pooling network strategy. Their study successfully detected single leaks in a simple single pipeline system, even in the presence of noise with a signal-to-noise ratio (SNR) as low as 0 dB. The approach was validated through numerical experiments. In the following year, Bohorquez et al. [37] used time-domain pressure head signals to train both an Artificial Neural Network (ANN) and a 1-D Convolutional Neural Network (1D-CNN) in a noise-free environment, aiming for a single leak detection and recognizing sudden area changes in a single pipeline. This method was only validated numerically. In 2021, Bohorquez et al. [38] presented another paper wherein the authors trained ANNs using burst-induced transient signals for the detection and localization of single burst occurrences in a single pipeline system. Their method was validated using both numerical simulations and laboratory experiments. In the same year, Liao et al. [39] adopted resonant frequency responses to train a multi-layer dense-net (ANN) algorithm, which could detect a single leak in large-scale pipeline networks, with validation conducted through numerical dataset. Recently, in 2022, Ayati et al. [40] classified leaks in pipeline networks by employing resonant frequency responses and Neighborhood Component Analysis (NCA). They also focused on single leak detection and validated their approach both numerically and in laboratory experiments.

In recent years, hybrid approaches, which combine physics-based models with data-driven techniques, have also shown promising results. In 2022, Waqar et al. [41] harnessed the capabilities of Physics-Informed Neural Networks (PINNs), integrating them for the first time with the principles of hydraulic transients in pipeline systems. The model was informed by the 1-D wave equation to predict the transient-wave propagation in a single pipeline. The study conducted a detailed sensitivity analysis regarding the architecture of the neural network and the nature of the input training data, underpinned by numerical validation. In parallel, Ye et al. [42] also employed a PINN, informed by the water-hammer equations with frictional and visco-elastic damping considerations. In addition to numerical exploration, the authors validated their findings through laboratory experiments, providing the efficacy of their findings.

These studies collectively underscore the progressive role of ML in enhancing the precision and reliability of leak detection in pipeline systems, showcasing a blend of diverse ML algorithms with remarkable performances. However, it is found that the majority of research efforts have concentrated on detecting a single leak in a given pipeline. Contrarily, the possibility of multiple simultaneous leaks within a pipeline system cannot be overlooked, especially considering that the occurrence of one leak may compromise the structural integrity of the system, thereby increasing the likelihood of subsequent breaches [43]. To address these limitations, this paper makes the following key contributions:

1. We propose a novel domain-guided machine learning-based framework specifically designed for the simultaneous detection and localization of multiple leaks in pressurized pipelines, overcoming the limitations of existing methods that primarily focus on detecting single leaks.
2. The framework incorporates modeling uncertainties, such as valve closure time, pre-transient pressure head, and flow rate variations, through a model refinement approach—an aspect not extensively addressed in previous studies.

3. The proposed method is validated using both comprehensive numerical datasets and experimental data from a Viscoelastic pipe system, demonstrating robustness and reliability in real-world applications.
4. The method shows computational efficiency, with baseline model training completed in approximately 2.5 h, and it adapts well to complex pipeline configurations, including series and tree-type networks.

These contributions offer a robust solution for multiple leak detection and pave the way for further exploration of complex systems and data-driven ML models to tackle increasingly challenging leak detection scenarios.

The remainder of this paper is organized as follows: Section 2 outlines the problem statement. Section 3 details the proposed framework and the flowchart for its implementation. Section 4 describes the numerical tests conducted to assess the proposed framework, including the integration of modeling uncertainties through model refinement techniques. Section 5 experimentally validates the method using distinct laboratory datasets. Section 6 presents the potential application of the proposed framework to pipeline networks. Finally, Section 7 presents the conclusions.

2. Problem statement

In practical applications, water pipeline systems are often subjected to significant levels of noise, originating from a variety of sources, including flow turbulence, mechanical vibrations induced by pumps and flow regulation devices, and fluctuations in consumer usage and demand [29,44]. Additionally, the environmental context of the pipelines – whether they are above ground or buried, and their proximity to highways, railway tracks, or other infrastructure – further contributes to noise characteristics. As such, the measured transient signals are inevitably tainted with noise. Let $h_m(t, s^L, x^L, n^L)$ be the measured transient response of the system, which can be approximated as:

$$h_m(t, s^L, x^L, n^L) = h_c(t, s^L, x^L, n^L) + \epsilon(t), \quad (1)$$

where s^L denotes the leak size, x^L is the leak location, n^L is the number of leaks, t denotes the time vector, $\epsilon(t)$ is the noise, and h_c is the predicted transient response and is denoted as

$$h_c(x^L, s^L, n^L) = \text{ForwardModel}(x^L, s^L, n^L) \quad (2)$$

Within this framework, the ForwardModel can be an explicit/implicit time-domain model, leveraging techniques like the Method of Characteristics or finite volume methods [9]. Alternatively, it could be a frequency domain-based model, utilizing either frequency response functions or impedance-based models. The challenge lies in extracting the leak information, namely s^L , x^L , and n^L , from the noisy signal $h_m(t)$. This task is significantly influenced by the signal-to-noise ratio (SNR). For leakage detection problem, the SNR is defined as [13,29]

$$\text{SNR} = 20 \log_{10} (\Delta h^L / \sigma_{\text{noise}}) \quad (3)$$

where Δh^L is average pressure head difference caused by the leak, and σ_{noise} is the standard deviation of noise.

Traditionally, retrieving defect information from noisy measurements is approached as an inverse problem. This process involves estimating the defect characteristics from observed data, typically governed by the equation:

$$\{\hat{x}^L, \hat{s}^L, \hat{n}^L\} = \text{InverseModel}(h_m(t)) \quad (4)$$

In this equation, $\{\hat{x}^L, \hat{s}^L, \hat{n}^L\}$ represent the estimated leak characteristics. The InverseModel function encapsulates the optimization approach to infer defect information from the observed data. However, this approach is inherently ill-posed due to the intrinsic noise in the measured signals. As such, the accuracy and reliability of the solutions derived from this model are heavily dependent on the algorithm employed [10,45–47]. In many cases, the presence of noise can lead to ambiguous or multiple solutions, resulting in false alarms [10,12].

In the literature, the use of a machine learning algorithm (e.g., ANN, CNN or NCA, as described in the previous section) as a surrogate inverse model is proposed to map noisy measurements to specific defect information. The underlying concept is mathematically represented as follows:

$$\{\hat{x}^L, \hat{s}^L\} = \text{ML}(h_m(t, x^L, s^L, n^L)) \quad (5)$$

where, ML denotes an arbitrary machine learning model, and $\{\hat{x}^L, \hat{s}^L\}$ symbolize the defect characteristics inferred from the hydraulic measurements $h_m(t, x^L, s^L)$. The current approach to solve Eq. (5) presents distinct challenges such as:

1. In the current framework, where leak parameters are directly estimated as outputs, an ML model trained for a single leak is predisposed to detect only one leak, whereas a model trained for two leaks will necessarily identify exactly two, regardless of the actual conditions. This reveals a limitation of the existing framework's adaptability to varying numbers of actual leaks. Such a limitation originates from the ML model's architecture, designed to detect only a specific number of leaks, n_L . As a result, the model inherently biases its predictions towards the number of leaks for which it was initially trained. In practical scenarios, where the actual number of leaks is undetermined, this bias leads to inaccurate predictions.
2. Adjusting the model to predict an alternate number of leaks would require changing the dimension of the output vector within the ML architecture. However, the architecture imposes strict limitations on the dimensions of the output vector, making such adjustments impractical. This inflexibility hinders the ability to fine-tune models for accurately detecting multiple leaks after their initial training.

This framework of prediction, driven by the model's training constraints, raises significant concerns about the accuracy and practicality of the detection method. It also highlights the need for developing more adaptable and accurate leak detection framework strategies that can effectively manage the dynamic nature of pipeline systems.

3. Proposed domain-guided framework to detect multiple leaks

In this paper, we introduce a novel framework for detecting multiple leaks simultaneously using a single ML model, without constraining the target vector based on the number of leaks. Drawing inspiration from the output functions of matched-field processing (MFP, [29]) and time reversal method (TRM, [13]), we define a domain-guided output function $\mathcal{L}(x^L, s^L, n^L)$, which maintains a fixed dimension regardless of the number of leaks. In this function, each potential leak point is represented by a density-like lobe, as demonstrated in Fig. 1. This lobe, centered at the leak's location, has its width proportional to the minimum wavelength of the transient signal injected into the system, akin to the principles found in MFP and TRM. Mathematically, this output function is described as

$$\mathcal{L}(x^L, s^L, n^L) = \sum_{i=1}^{n^L} s_i^L \cdot \exp\left(-\frac{(x - x_i^L)^2}{2\sigma^2}\right) \quad (6)$$

where $\mathcal{L}(x^L, s^L, n^L)$ is the leak function, x_i^L denotes a peak in the function, indicating the location of the i th leak, and s_i^L is the peak magnitude of the lobe, representing the i th leak size. The term $\sigma = \lambda / (2\sqrt{2 \ln(20)})$ corresponds to the effective width of the lobe and is defined based on the minimum wavelength λ of the injected wave, as highlighted in Fig. 1. As such, the former framework in Eq. (5) is redefined as

$$\hat{\mathcal{L}}(x^L, s^L, n^L) = \text{ML} (h_m(t, x^L, s^L, n^L)) \quad (7)$$

where $\hat{\mathcal{L}}(\hat{x}^L, \hat{s}^L, \hat{n}^L)$ is an estimate of $\mathcal{L}(x^L, s^L, n^L)$. This framework specifically addresses the aforementioned limitations as follows:

1. The ML model predicts an output function that indirectly represents leak properties, allowing for training across different numbers of leaks. This flexibility mitigates the issue of discrepancies between the number of leaks in training and actual scenarios, as long as the actual number of leaks is fewer than or equal to what the model was trained on. This limitation is not inherent to the proposed framework but stems from a well-known constraint in ML models: their inability to generalize beyond the scope of their training data.
2. The number of elements in the leak function, denoted as $\mathcal{L}$, is determined by the discretization of the variable x , rather than by the number of leaks. This design feature allows the model to adapt without the need to develop separate models for each leak scenario. Moreover, consistent output size simplifies fine-tuning, enhances scalability, and improves efficiency, enabling the model to handle varying datasets and uncertainties more effectively.
3. In addition, since the proposed method utilizes the full waveform, its performance is not hindered by blind spots – such as zero-pressure points in resonant modes – where leaks may go undetected [48,49]. As such, given that the model is trained on a comprehensive range of potential leak locations, it enables the detection of leak signatures across all areas, ensuring robust performance even in scenarios where traditional spectral-based methods face limitations [49].

3.1. Methodology

3.1.1. General

The methodology flowchart is outlined in Fig. 2 and is divided into three main blocks, each of which is detailed in subsequent sections.

3.1.2. Block 1: Development of the hydraulic twin model

The initial phase in this block involves selecting a pipeline system and designing experiments. This includes identifying access locations, understanding pipeline geometry, and defining operational conditions such as flow rates, pressure heads, consumer demands, locations for transient generation, and sensor installation. The next step is to develop and calibrate a numerical model—a virtual and hydraulic transient twin of the real system.

To calibrate the hydraulic model using the experimental dataset, if data is available, model parameters such as wave speed, damping factors, excited waveforms, bandwidth, and amplitude are iteratively tuned. This tuning process involves adjusting these parameters based on the comparison between simulated pressure head signals and the measurements recorded from sensors installed in the real pipeline system. The adjustments continue until the simulated results closely replicate the observed data, ensuring the numerical model accurately captures the dynamics of the physical system. In cases where experimental data is not available, educated estimates and engineering expertise are employed to tune the model based on known pipeline characteristics, such as material, diameter, and layout, as well as historical performance data. This approach provides a reliable baseline, ensuring the model reflects the expected behavior of the system.

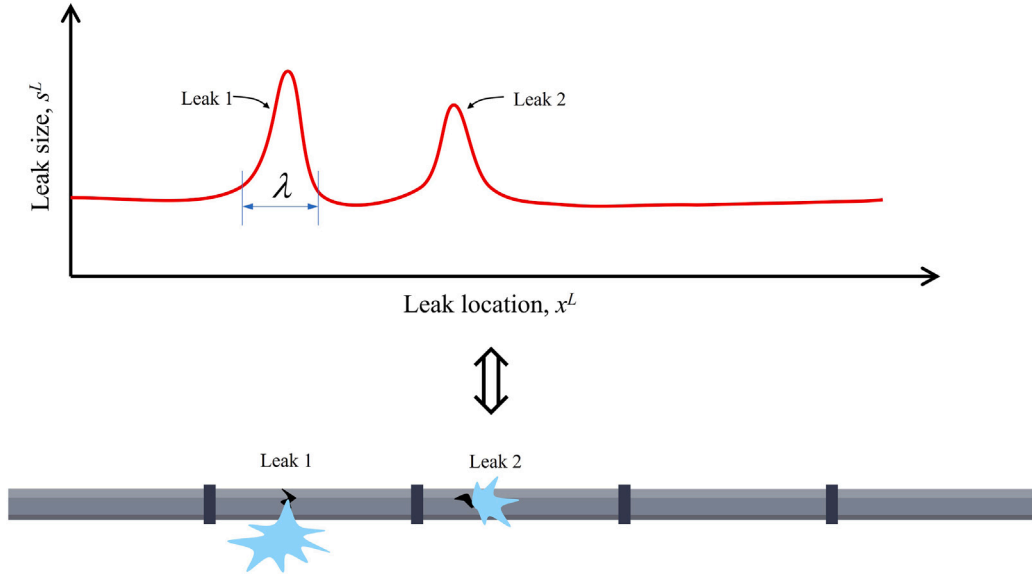


Fig. 1. Representation of multiple pipeline leaks as a probability density function where the peak's amplitude indicates the leak size, the peak's position corresponds to the leak location, and the standard deviation is related to the wavelength of the injected transient signal.

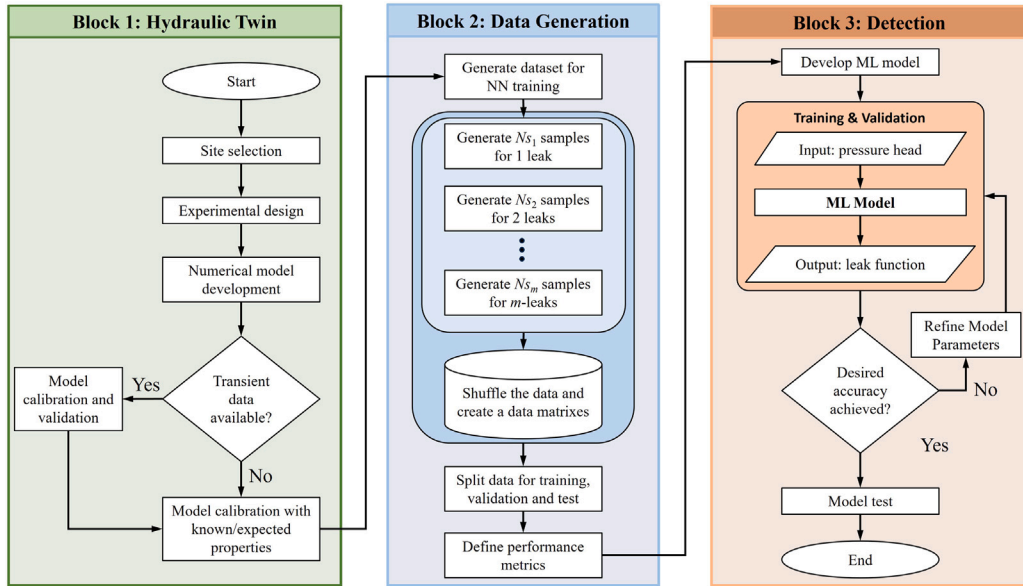


Fig. 2. Flowchart illustrating the methodology for developing and validating a machine learning model for leak detection.

3.1.3. Block 2: Data generation for ML model training

Data generation for training the ML model is executed using the calibrated hydraulic transient twin model. This process involves simulating transient responses for varying numbers and sizes of leaks. The number of samples per leak number depends on the computational complexity and the hydraulic engineer's discretion. We recommend starting with simulations for small SNR values, as smaller leaks are more challenging to detect. Leak locations should vary between λ and $L - \lambda$, where L is the pipe length and λ is the wavelength of the injected transient signal. For post-simulation, the defect information (leak size, location, and number) for each sample is transformed into a leak function, as described in Eq. (6). This forms the basis of our proposed framework, where the ML model maps the measured transient response to the corresponding leak function, which could have 1, 2, or n^L peaks.

3.1.4. Block 3: Development of the machine learning model

In this final block, a regression-based ML model is developed. The input to this model is the measured pressure head ($h_m(t, x^L, s^L, n^L)$), while the output is the leak function ($\mathcal{L}(x^L, s^L, n^L)$). Our paper utilizes an Artificial Neural Network (ANN) as a surrogate model for this mapping, although the process is applicable to other models. The simulated data is split into training, validation, and testing segments. For testing purposes, a new dataset can be generated from the calibrated hydraulic model. Moreover, the training dataset with different number of leaks can be simulated and used simultaneously for ML model training.

3.1.5. Performance indicators

To evaluate the effectiveness of the regression model, several parameters including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R^2 (R-square), and Pearson Correlation Coefficient (PCC) are utilized [50]. Ideally, MSE, RMSE, and MAE are error metrics that should be minimized; thus, lower values directly correlate to better model performance. There are no upper bounds for these three indicators; they can range from 0 to infinity, where 0 represents a perfect fit to the data. On the other hand, PCC and R^2 are bounded metrics where higher values indicate better performance. The PCC has a range from -1 to 1 , with 1 being a perfect positive linear correlation, 0 indicating no correlation, and -1 signifying a perfect negative linear correlation. The R^2 metric is bounded between 0 and 1 , where 1 indicates that the model explains all the variability of the target variable around its mean, and 0 implies that the model fails to capture any of the variance of the target variable.

4. Numerical analysis

4.1. Training data set

Consider a reservoir-pipeline-valve system comprising a 500 m long pipe with an internal diameter of 150 mm and a wall thickness of 5 mm. The pipeline is modeled as an elastic structure characterized by a Young's modulus of 170 GPa and a Darcy-Weisbach friction factor of 0.015. Consequently, the transient wave speed, denoted as a_0 , is 1.25 km/s. The system operates with a pre-transient flow rate and pressure head of 2 LPS and 40 m at the downstream valve, respectively. This setup results in a Joukowski pressure head (H_J) of approximately 14.4 m above the steady-state pressure head.

For the simulation, we consider 3 wave periods, equating to a total time of $T = 12 \times L/a_0 = 4.796$ seconds. A sampling frequency of $f_s = 250$ Hz is utilized, yielding 1200 data points for each transient signal. The spatial resolution of the simulation is $\Delta x = 4.9783$ m, allowing for a minimum distance of Δx between two potential leaks. The transient wave is initiated by a valve maneuver with a time of $T_c = 0.03$ s, resulting in a minimum wavelength $\lambda_{\min} = a_0 \times T_c = 37.3$ m for the injected transient wave. Consequently, the diffraction limit [51,52] is approximately $\lambda_{\min}/2 = 18.67$ m. The transient signals recorded at the valve location are utilized for the ML model development.

For leakage simulation, leak sizes and locations are randomly chosen from a uniform distribution. Leak sizes range between $(0, 0.01A_p)$, where A_p is the cross-sectional area of the pipe and leak locations are selected within the bounds of $[\lambda, L - \lambda]$. The datasets include scenarios of 1, 2, and 3 leaks, each simulated 2000 times, resulting in a total of 6000 samples. Each sample comprises 1200 data representing the pressure head. Every pressure head signal is then corrupted by adding the noise. The simulated noise is sampled from a Gaussian distribution with 0 mean and standard deviation of σ_{noise} , calculated from Eq. (3) as

$$\sigma_{\text{noise}} = \Delta h^L \times 10^{-\text{SNR}/20} \quad (8)$$

where Δh_L is the leak-induced pressure head difference, and SNR = 10 dB is selected for illustration purposes. Fig. 3 provides illustrative examples of transient signals for scenarios involving single, double, and triple leaks, each randomly selected from the dataset. In these plots, the horizontal axes represent the time and vertical axes denote the perturbed pressure head in response to the valve maneuver.

4.1.1. Dataset pre-processing

Data preprocessing is a critical aspect of our framework. The transformation of pressure head signals, which serve as the input to the ML model, is achieved by converting step-responses into pulse responses. This conversion involves subtracting a delayed copy of the original signals, a method outlined in [53]. Additionally, the leak properties, constituting the output of the ML model, are represented through a leak function as defined in Eq. (6). Illustrative examples of these transformations are presented in Figs. 4 and 5. Fig. 4 displays the conversion of signals from step to pulse responses, while Fig. 5 depicts the leak function for different leak scenarios (one, two, and three leaks in the top, middle, and bottom rows, respectively). In the latter plots, the horizontal axes are normalized against the pipe length, and the vertical axes are normalized in relation to the pipe's cross-sectional area.

The rationale behind transforming the signals into pulse responses is that pulses reflecting from leaks correlate more strongly with their corresponding leak function than step-formatted reflections. This hypothesis was substantiated through the training of the optimized neural network. It was observed that using pulse-type input signals yielded higher accuracy compared to step-like signals. For brevity, the detailed analysis supporting this observation is not included here.

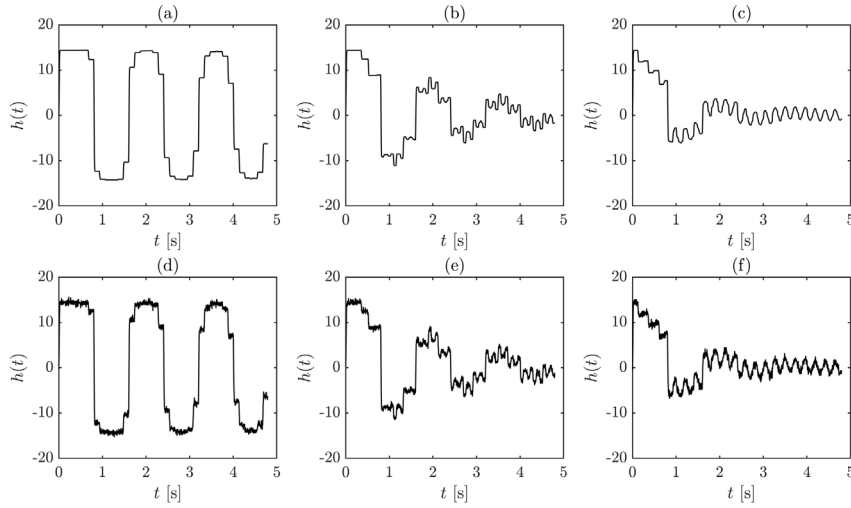


Fig. 3. Examples of numerically simulated signals without noise (top row) and with noise (bottom row), respectively for (a,d) one leak, (b,e) two leaks, and (c,f) three leaks.

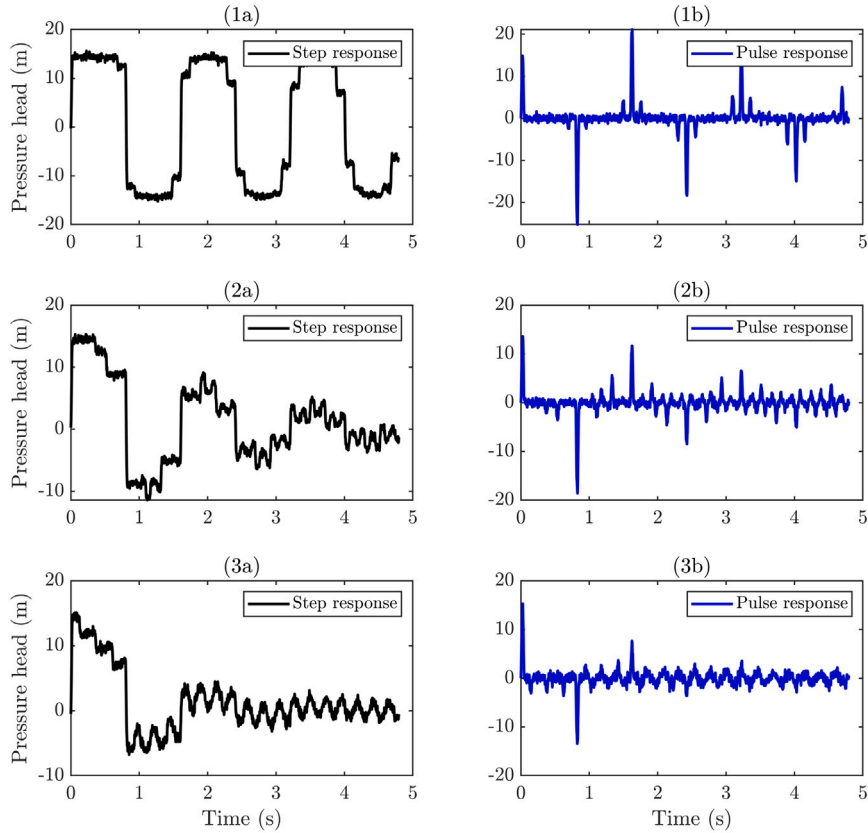


Fig. 4. Comparison of pressure-time signals: Step responses (left column) versus pulse responses (right column). Each row represents different leak scenarios with one, two and three leaks in first, second, and third row, respectively.

4.2. NN training, validation and testing

In this paper, a feedforward neural network, a type of Artificial Neural Network (ANN), is employed. The network's structure and hyperparameters have been carefully optimized using a trial-and-error approach. This method ensures the fine-tuning of the

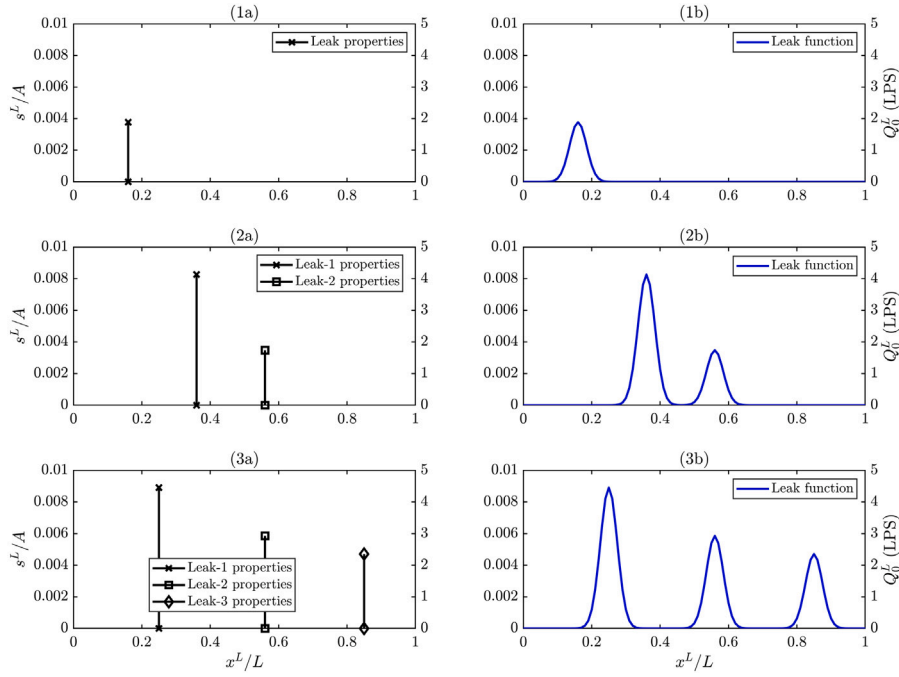


Fig. 5. Examples of numerically simulated leak properties in the hydraulic model for one, two and three leaks in 1(a), 2(a), and 3(a), and their corresponding leak functions in 1(b), 2(b), and 3(b), respectively. The left and right vertical axes denote the ratio of the leak size with respect to the pipe cross-sectional area and steady-state leakage flow rate, respectively. Note that, the leak sizes ranges between 0 to 1% of the cross-sectional area of the pipe, i.e., $[s_{\min}^L, s_{\max}^L] = [0, 0.01] \times A$.

Table 1
Neural network hyperparameters.

Parameter	Value
Hidden layer sizes	400, 400, 300, 300, 200
Learning rate	0.01
Momentum constant	0.9
Maximum epochs	30,000
Performance goal	1×10^{-5}
Validation fail count	1000

model to best fit the specific requirements of our dataset. The optimized model network consists of five hidden layers with varying numbers of neurons, designed to capture complex data patterns. The layers are configured with 400, 400, 300, 300, and 200 neurons, respectively. Data splitting is strategically done with 70% for training, 20% for validation, and 10% for testing, to prevent overfitting and ensure robust model evaluation. The training employs scaled conjugate gradient backpropagation (`trainscg`) optimizer, known for its low computational cost. The hidden layers use the tan-sigmoid function (`tansig`) as the activation function, effectively mapping input values between -1 and 1 . The hyperparameters of the network are detailed in Table 1.

Fig. 6 illustrates three key aspects of the training dynamics for the ANN model under study. Subfigure (a) presents the Mean Squared Error (MSE), which serves as the loss function for the ANN, plotted over the number of epochs. The MSE tracks the average of the squares of the errors, i.e., the differences between the predicted and actual values. The training curve shows a decreasing trend, indicating learning and model improvement over time. Notably, the MSE for the validation set initially decreases alongside the training set, suggesting that the model is generalizing well. The dashed ‘Best’ line marks the epoch at which the validation MSE is at its lowest, a potential candidate for model selection. The ‘Goal’ line represents the pre-defined performance goal, which the training process strives to achieve or surpass. Subfigure (b) depicts the gradient of the loss function with respect to the epochs. The gradient’s magnitude is used during training to update the model weights, with a descending trend reflecting the model’s convergence towards a minimum in the loss function. Subfigure (c) displays the validation fails over epochs, which counts the number of consecutive epochs where the validation performance has not improved. The model’s training continues until this count reaches the pre-set maximum of 1000, as denoted by the threshold line. Spikes in the validation fail count can be indicative of the model’s difficulty in further reducing validation error, potentially due to the complexity of the feature space or noise within the data. Once the validation fail count reaches 1000, training is halted to prevent overfitting, as the lack of improvement in validation performance could imply that the model has learned the training data too closely. Overall, these graphs collectively provide a comprehensive view of the training and validation process, demonstrating the ANN’s ability to learn from the data and the effectiveness of the training algorithm in minimizing error while avoiding overfitting.

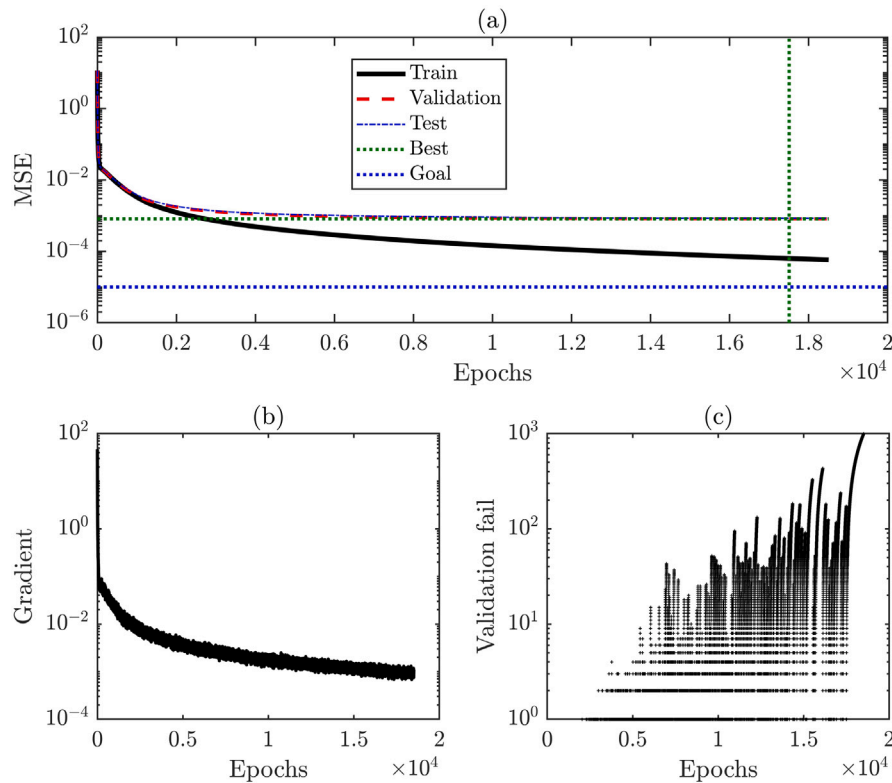


Fig. 6. Training performance metrics for the ANN model. (a) Mean Squared Error (MSE) loss function over epochs for training, validation, and test sets. The 'Best' line indicates the epoch with the lowest validation MSE, and the 'Goal' line represents the targeted MSE threshold. (b) Gradient of the loss function over epochs, demonstrating the optimization process. (c) Validation fails over epochs, with a threshold set to 1000.

To summarize, the NN model was trained using a dataset consisting of 6000 samples, each with 1200 features, and the output layer corresponding to 6000 samples with 101 features each. The network architecture comprised of an input layer, multiple hidden layers, and an output layer, amounting to a total of 972,101 trainable parameters. This level of complexity in the model structure allows for intricate pattern recognition and was managed within a reasonable training duration of 2 h and 36 min on a 12th Generation Intel Core i9-12900 CPU with 2.4 GHz and 32 GB of RAM. The training of the ANN model was conducted using MATLAB 2022 and was completed on a CPU without the aid of GPU acceleration, highlighting the model's efficiency on standard hardware configurations. Moreover, the convergence of training was based on the model achieving a specified level of accuracy as described earlier, ensuring the robustness of the predictions against the diverse range of data presented.

Fig. 7 offers a comprehensive view of the ANN model's predictive performance across different datasets. Subfigure (a) illustrates the model's performance on the training set, with an exceptional degree of fit as evidenced by the high R^2 value, suggesting that the model has learned the training data effectively. The proximity of the data points to the 1:1 line indicates accurate predictions relative to the observed data. Subfigure (b) represents the validation set, which is used to tune the model's hyperparameters. The plot shows a slightly wider dispersion of points compared to the training set, yet maintains a high R^2 value, indicating the model's good generalizability to unseen data. Subfigure (c) displays the testing set, which assesses the model's performance on entirely new data. While there is a further increase in the spread of data points around the regression line, the R^2 , RMSE, MSE, and MAE values confirm that the model's predictive accuracy remains high. The PCC across all datasets remains robust, highlighting the model's consistent ability to predict leak properties accurately. These results collectively demonstrate the ANN model's robustness and reliability in leak detection, from learning during training to generalizing well in validation and testing phases.

It is essential to note that while regression plots are informative, they do not always provide a complete picture of a model's predictive performance, especially when considering the full scope of leak properties (numbers, sizes and locations). As such, reliance on regression plots alone may not appropriately depict the model's capabilities. Therefore, a domain-guided approach to performance assessment is employed, as depicted in Fig. 8. Fig. 8 elucidates the ANN model's ability to predict leak properties. This is achieved by plotting the normalized leak size against the normalized leak location for each scenario. This approach takes into account the intricacies of leak size, numbers, and location within the normalized bounds, providing a more nuanced understanding of the model's effectiveness. As seen across the nine plots, the model exhibits consistent precision in predicting leak properties for a single leak. As the number of leaks increases, the complexity of the prediction task also rises, reflected by the slight divergence between the true and predicted values. However, even in the presence of multiple leaks, the model maintains a noteworthy level of

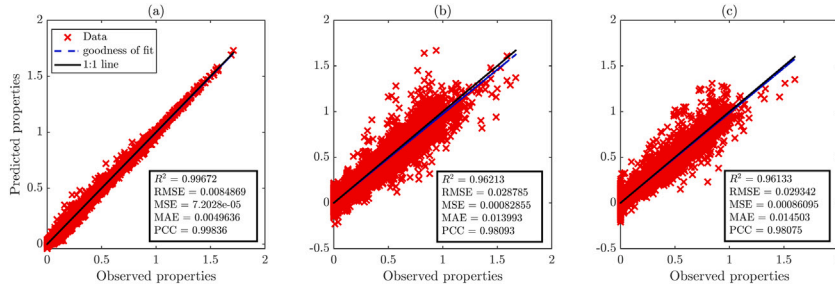


Fig. 7. Scatter plots of the ANN model's performance for predicting leak properties across different datasets: (a) training, (b) validation, and (c) testing. The observed properties are plotted along the x-axis, and the predicted properties are on the y-axis. The red crosses represent individual predictions, the blue dashed line shows the regression line indicating the goodness of fit, and the solid black line depicts the ideal 1:1 relationship between observed and predicted values. Insets in each sub-figure display the model's performance metrics: the coefficient of determination (R^2), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), Mean Absolute Error (MAE), and Pearson Correlation Coefficient (PCC).

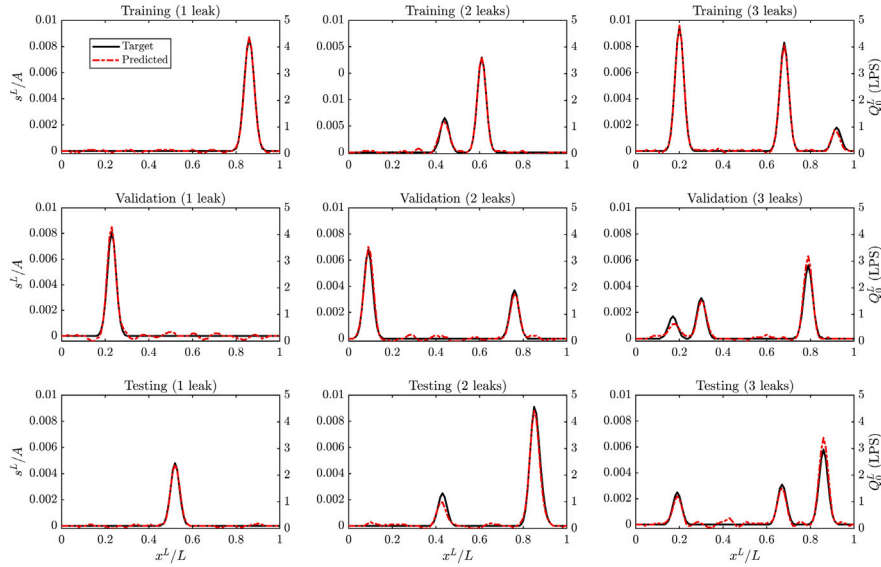


Fig. 8. Comparative plots of true versus predicted leak properties for the ANN model. Each row corresponds to a different dataset: training, validation, and testing; while each column represents the number of leaks: one, two, and three. The normalized leak size s^L/A is plotted against the normalized leak location x^L/L . The solid black line indicates the true values, and the dashed red line represents the ANN predictions.

accuracy, as indicated by the close alignment of the predicted values to the true leak characteristics. This comprehensive evaluation method, which combines regression analysis, offers a robust framework for validating the ANN model's performance in detecting and quantifying leaks in pipeline systems.

4.3. Parameters uncertainty and model refinement

Conducting repeatable transient experiments in hydraulic systems poses unique challenges, particularly when relying on manual valve operations. Such manual interventions often result in variability in valve closure times, diverging from the conditions under which a machine learning model is usually trained. This variability is a critical aspect, as it significantly influences the characteristics of the transient waves such as the bandwidth, waveform, and wavelength. To address this, we have adapted our pre-trained model, designated as BaseModel, using refinement techniques. The initial phase of adaptation involves retraining BaseModel on a dataset that reflects a range of valve closure times ($T_c \in [0.02, 0.08]$ seconds), capturing the realistic operational variations. The refined model is referred to as AdaptedModel – 1. The training performance curves (i.e., loss function, gradient of the loss function and validation fails) are provided in Fig. A.19 (in Appendix). The performance metrics are summarized in Table 2. The efficacy of the refined model in predicting leak functions is depicted in Fig. 9, where the influence of valve closure time (T_c) is apparent. The width of the leak signature lobe in each subplot correlates with the wavelength of the induced signal, which is integral to the leak function computation as stipulated in Eq. (6).

In the subsequent refinement stage, AdaptedModel – 1 undergoes additional enhancement to account for variations in the pre-transient flow rate at the valve location, symbolized as $Q_{0,v}$, along with the valve closure time ($T_c \in [0.02, 0.08]$ seconds). The

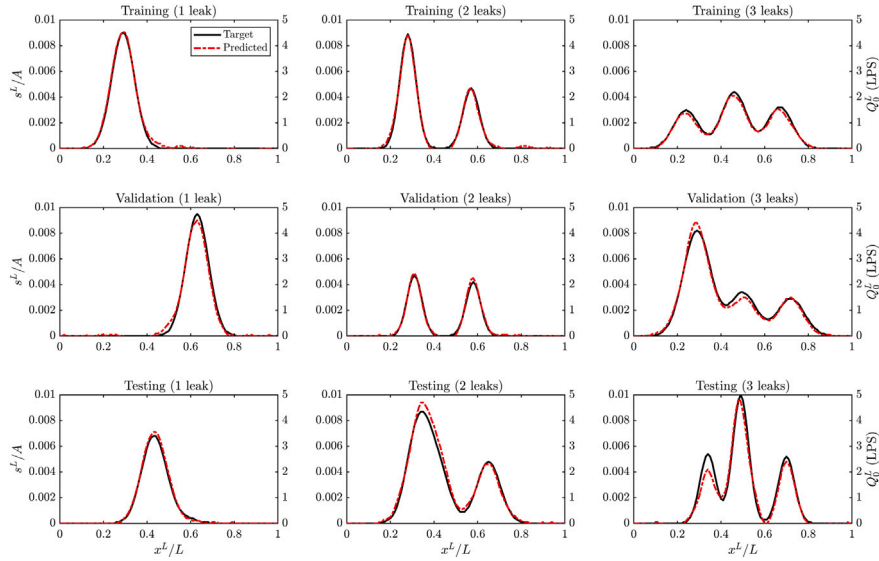


Fig. 9. Comparative plots of true versus predicted leak properties for the AdaptedModel – 1. Each row corresponds to a different dataset: training, validation, and testing; while each column represents the number of leaks: one, two, and three. The normalized leak size s^L/A is plotted against the normalized leak location x^L/L . The solid black line indicates the true values, and the dashed red line represents the ANN predictions. The role of valve time closure (T_v) is indicative in terms of the width of the localizing lobe in each sub-plot as it is dictated by the wavelength of the injected signal which is directly used in Eq. (6) to compute the leak function.

integration of $Q_{0,v}$ is crucial, as it dictates the amplitude of the incident wave through Joukowski's pressure head, defined by $H_j = (a_0 Q_{0,v})/(gA)$, where a_0 represents the wave speed, g is the gravitational acceleration, and A is the cross-sectional area of the pipe. Notably, the reflection coefficients from potential defects are directly proportional to Joukowski's pressure head, making it a vital factor in accurately sizing defects. Omitting this variability could lead to significant errors in estimating defect size. To address this, the flow rate is varied within the range of 1 to 3 liters per second, and an enriched dataset encompassing scenarios of 1, 2, and 3 leaks is constructed, with each scenario comprising 2000 samples. This dataset serves to fine-tune AdaptedModel – 1, now relabeled as AdaptedModel – 2. The training performance curves for this enhanced model are exhibited in Fig. A.20, detailed in Appendix. Table 2 provides a detailed account of the improvements in performance metrics for AdaptedModel – 2. The efficacy of this model in predicting the leak function is presented in Fig. 10, which illustrates the model's performance across the training, validation, and testing stages, sequentially presented in the top, middle, and bottom rows. While there are slight discrepancies in leak size prediction, the precision in leak localization is maintained with remarkable accuracy, and no extraneous side lobes are evident. This fortifies the model's dependability in consistently identifying leak characteristics, including size, location, and quantity.

Subsequently, AdaptedModel – 2 advances to tackle scenarios involving four leaks. To facilitate this, we simulate an augmented dataset comprising 2000 samples, each characterized by four leaks, which serves as the new training ground for the model. The updated model, designated as AdaptedModel – 3, is adept at capturing the variability across three primary parameters: valve closure time, pre-transient flow rate, and the number of leaks—now extending up to four. The training performance of AdaptedModel – 3, delineated by its learning curves, is showcased in Fig. A.21 (refer to Appendix) and the values of various performance metrics are provided in Table 2, reflecting the model's ability to adapt to both valve closure time and flowrate variability. Furthermore, the model's predictive accuracy concerning the leak function is demonstrated in Fig. 11. This figure selectively exhibits three instances from each phase – training, validation, and testing – to visually convey the model's proficiency. Notably, AdaptedModel – 3 consistently identifies all four leaks with remarkable precision in both size and location.

Fig. 12 visualizes the performance metrics detailed in Table 2, offering a graphical representation of the machine learning models' proficiency throughout the training, validation, and testing stages. In subfigures (a) through (e), the error metrics – MSE, RMSE, and MAE – exhibit an ascending trend, and the fit quality indicators – R-squared and PCC – show a descending trend as the BaseModel evolves through AdaptedModel – 1 to AdaptedModel – 3. This trend reflects the models' confrontation with increasing variability in the input data, including uncertainties in critical parameters such as valve closure time, pre-transient flow rate, and the number of undetermined leaks. While an escalation in error through the stages of validation and testing relative to training is observed, suggesting no over-fitting, it indicates the model's increasing challenges in capturing the complexity of the data. This increase is a predictable consequence of exposing the model to broader operational dynamics and serves as an impetus for further enhancement. The incorporation of a more diverse training dataset stands as a viable approach to refine the model's performance, enabling it to better generalize across the spectrum of variability and uncertainty inherent in real-world applications.

The examples presented herein serve to underscore the effectiveness of the proposed framework, which adeptly accommodates variability in both physical parameters and unforeseen operational conditions. This adaptability is achieved through the addition of a

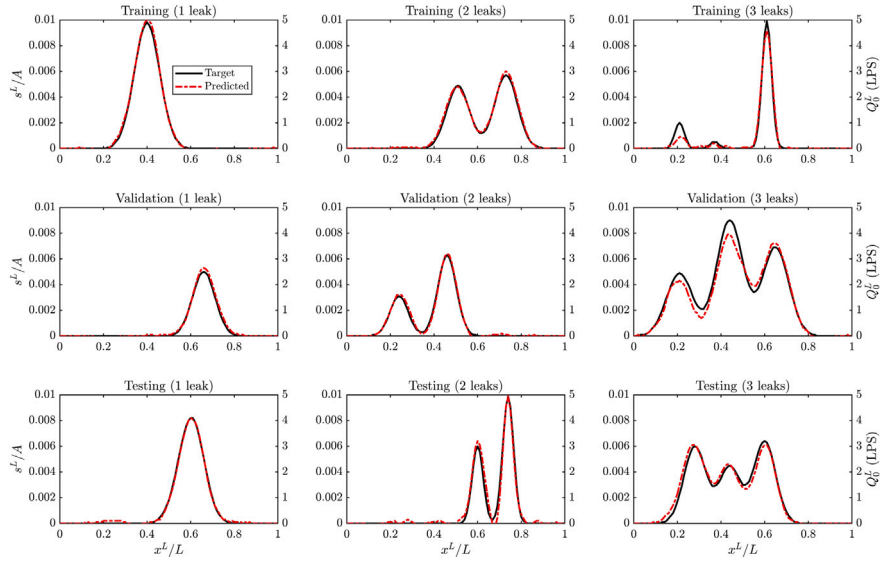


Fig. 10. Comparative plots of true versus predicted leak properties for the AdaptedModel-2. Each row corresponds to a different dataset: training, validation, and testing; while each column represents the number of leaks: one, two, and three. The normalized leak size s^L/A is plotted against the normalized leak location x^L/L . The solid black line indicates the true values, and the dashed red line represents the ANN predictions. The role of valve time closure (T_c) is indicative in terms of the width of the localizing lobe in each sub-plot as it is dictated by the wavelength of the injected signal which is directly used in Eq. (6) to compute the leak function.

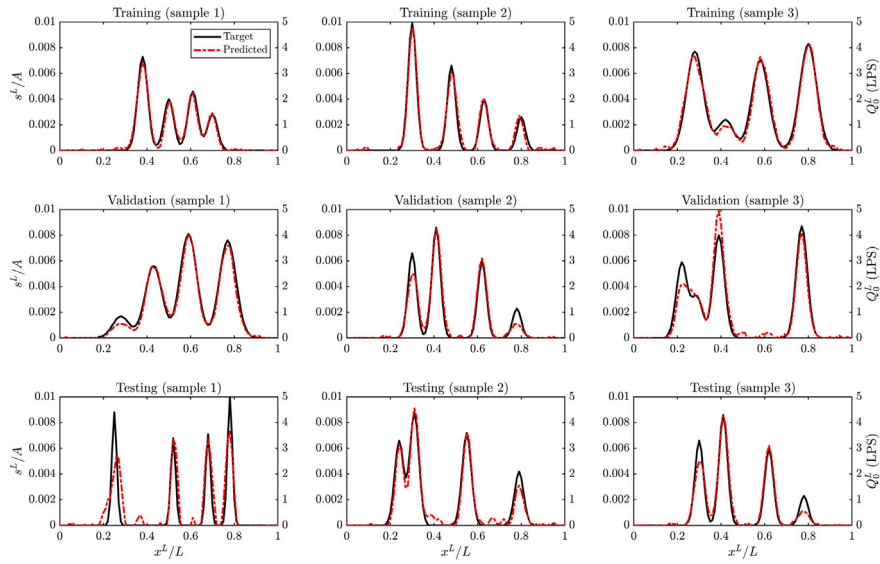


Fig. 11. Comparative plots of true versus predicted leak properties for the AdaptedModel-3. Each row corresponds to a different dataset: training, validation, and testing; while each column represents different samples. The normalized leak size s^L/A is plotted against the normalized leak location x^L/L . The solid black line indicates the true values, and the dashed red line represents the ANN predictions. The role of valve time closure (T_c) is indicative in terms of the width of the localizing lobe in each sub-plot as it is dictated by the wavelength of the injected signal which is directly used in Eq. (6) to compute the leak function.

relatively small training dataset, demonstrating that significant refinements can be realized in a remarkably short training duration—approximately 50 min for each stage of enhancement. Such illustrative advancements pave the way for future improvements, including the accommodation of wave speed variations, the mitigation of ambient noise, and the handling of scenarios involving an increased number of leaks. Crucially, our model maintains a consistent output dimensionality regardless of the number of leaks, distinguishing it from traditional approaches that require resizing the output vector to match the number of predicted leak sizes. This consistency is a testament to the potential of our methodology for practical field deployment, offering a paradigm shift in the use of machine learning for leak detection within hydraulic systems.

In what follows, the experimental validation of the proposed framework through a series of laboratory experiments is provided.

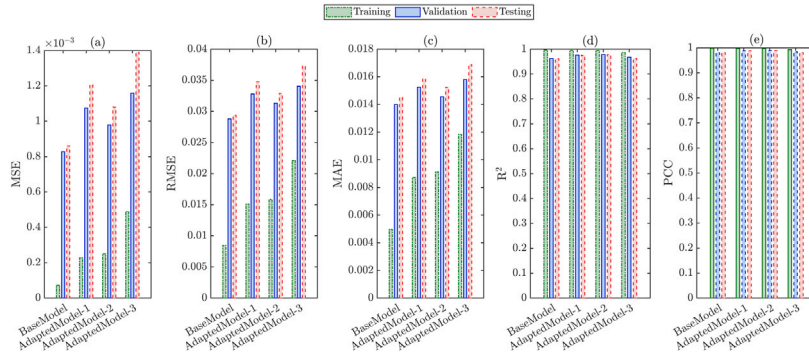


Fig. 12. Comparative evaluation of model performance metrics: This figure illustrates the progression of performance metrics across different models, from the BaseModel to AdaptedModel – 3. Subfigures (a) through (e) respectively depict MSE, RMSE, MAE, R-squared, and PCC, showcasing each model's performance during training (green), validation (blue), and testing (red) phases. The systematic reduction in error metrics (MSE, RMSE, and MAE) and concurrent increase in R-squared and PCC from BaseModel to AdaptedModel-3 highlight the efficacy of the refinement process in enhancing model accuracy and consistency across various stages of evaluation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Performance metrics of various models during training, validation and testing stages.

Model	Stage	R-squared	RMSE	MSE	MAE	PCC
BaseModel	Training	0.99672	0.008486	7.2×10^{-5}	0.00496	0.99836
	Validation	0.96216	0.028785	0.00082855	0.013993	0.98093
	Testing	0.96133	0.029342	0.00086095	0.014503	0.98075
AdaptedModel – 1	Training	0.99514	0.015099	0.00022797	0.0087127	0.99757
	Validation	0.97558	0.034736	0.0012066	0.015862	0.98773
	Testing	0.97634	0.032789	0.0010747	0.01525	0.9881
AdaptedModel – 2	Training	0.99451	0.015836	0.00025078	0.0091151	0.99725
	Validation	0.97806	0.031309	0.00098026	0.014559	0.98901
	Testing	0.97767	0.032868	0.0010803	0.015245	0.98883
AdaptedModel – 3	Training	0.98614	0.022106	0.00048867	0.011841	0.99304
	Validation	0.96784	0.034039	0.0011587	0.015799	0.98381
	Testing	0.96231	0.037285	0.0013901	0.016879	0.98102

5. Experimental validation

5.1. System description

The test rig is located at the Water Engineering Laboratory of the University of Perugia, Italy. The schematic of the experimental arrangement is delineated in Fig. 13. The configuration comprises an air vessel situated upstream, a horizontal pipeline made of High-Density Polyethylene (HDPE), and a pneumatic valve positioned at the downstream terminus. A submerged pump, placed within the recycling tank, ensures the stabilization of the pressure head within the air vessel. The pipeline's specifications include a length (L) of 188.89 m, an internal diameter (D) of 0.0933 m, a Darcy-Weisbach friction factor (f) calibrated at 0.0196, and an elastic modulus (E) of the pipe material registered at 1.47 GPa. The Viscoelastic (VE) characteristics of the pipe walls are incorporated via five Kelvin-Voigt model elements. The calibrated creep coefficients are delineated as $\vec{J} = [0.0657, 0.1054, 0.0091, 0.0261, 0.0746] \times 10^{-9} \text{ Pa}^{-1}$. The corresponding retardation times are $\vec{\tau} = [0.05, 0.5, 1.5, 5, 10] \text{ s}$. To facilitate leak simulations, two steel plates, each encompassing a hole, are installed at locations $x^{L1} = 83.19 \text{ m}$ and $x^{L2} = 110.53 \text{ m}$ along the pipeline.

Owing to spatial limitations within the laboratory, the pipeline assumes a helical configuration, integrating both straight and curved sections, each curved segment approximating an angular displacement of 180 degrees and a radius of curvature of approximately 1.5 m. The pipeline is predominantly suspended approximately 1 foot above ground level, barring a few meters adjacent to the upstream end. This elevation is supported by steel saddle clamps, which are affixed to a U-shaped hang-rail embedded within the ground.

5.1.1. Transient generation and data acquisition

Transients are generated by rapidly and completely closing the initially open valve (in about 35 ms) located at the downstream end. A piezoresistive transducer with a full scale of 0.4 MPa (2–4 bar) is placed at $x_m = 187.66 \text{ m}$ to measure the pressure head response of the system. National Instrument cDAQ-9188 data acquisition system with a sampling frequency of 2048 Hz is used to record the data.

We investigate four scenarios in the hydraulic system: Case 1 without any leaks; Case 2 with a single leak of 33.6 mm^2 at x^{L1} ; Case 3 with a single leak of 60 mm^2 at x^{L2} ; and Case 4, which is a composite scenario encompassing the leaks from both Case 2 and

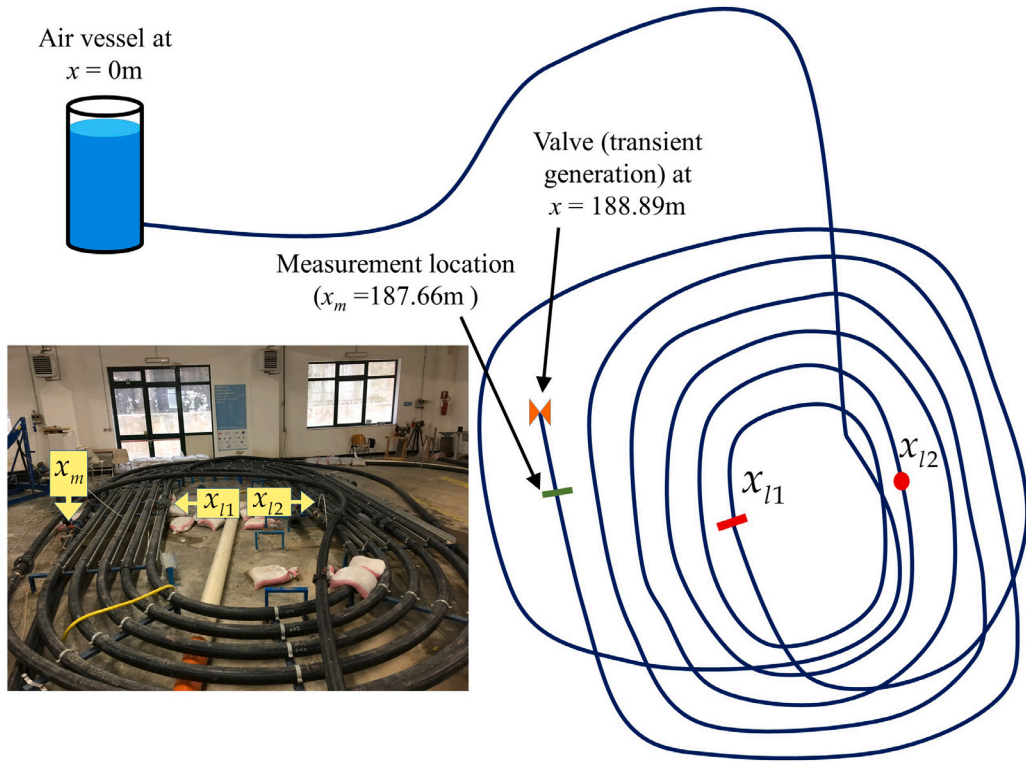


Fig. 13. Schematic of the experimental setup at Water Engineering Laboratory of the University of Perugia (UNIPG), Italy.

Case 3. Consequently, the experimental dataset includes two instances of a single leak and one instance of a two-leak configuration. The resulting pressure head signals are depicted in Fig. 14. A numerical model, with system characteristics outlined in Section 5.1, is calibrated against Case 1 (i.e., no leak case). This calibrated model then serves to synthesize the training data. A total of 4000 simulations are executed—2000 for a single leak and 2000 for two leaks. Leak characteristics within each simulation are randomly assigned based on a Uniform distribution, adhering to a leak size threshold ranging from 16.8 mm² to 72 mm² and leak locations distributed between λ and $L - \lambda$. Here, λ is determined by the maximum frequency component excited in the transient signal.

A feedforward neural network, identical in structure and hyperparameters to those used in the numerical examples, is deployed to develop the initial BaseModel. Model refinement is executed in a stepwise manner. First, the model is adapted to account for valve closure time variations between 0.02 to 0.04 s. Subsequent refinements incorporate pre-transient pressure head variations from 18 to 20 m, followed by flow rate adjustments between 1.4 to 1.8 liters per second. For each stage of refinement, 4000 samples are generated, evenly divided between single-leak and two-leak scenarios. The refined model, denoted as ExpModel, is then applied to predict leaks using experimental data, as depicted in Fig. 14. The experimental data is pre-processed for compatibility with the model's input format, which involves down-sampling from 2048 Hz to 250 Hz and truncating at 2.5 s to encapsulate one complete wave cycle.

The effectiveness of the ExpModel is demonstrated by the comparison of predicted and target leak functions, which is detailed in Fig. 15. The figure showcases a series of leak function predictions made by the ExpModel, compared against the target data for three distinct cases (Case 2, Case 3, and Case 4), as well as against the no-leak scenario (Case 1) for reference. In subfigure (a), the model accurately predicts the leak location and size, with the predicted function closely matching the target. Subfigure (b) reveals a slight overestimation in the predicted leak size and a broader width in the leak function, yet the relative error is only about 10%. Similarly, subfigure (c) displays a minor shift in the localization of the first leak; however, the localization error is less than $0.5\lambda_{\min}$, which is within the physical threshold for accurate predictions. The presence of additional lobes in the predicted functions can be attributed to small oscillations in the experimental signal—oscillations that are absent in the numerical dataset. Despite these nuances, the model's predictions align remarkably well with the target data, demonstrating its effectiveness in leak detection and localization.

6. Extension to pipeline networks

6.1. General

This section provides the extension procedure of the proposed framework to pipeline networks. The key to this extension is to organize the leak functions into a row or column vector. For example, consider a network consisting of NP pipes, where $\mathcal{L}_n$ (for

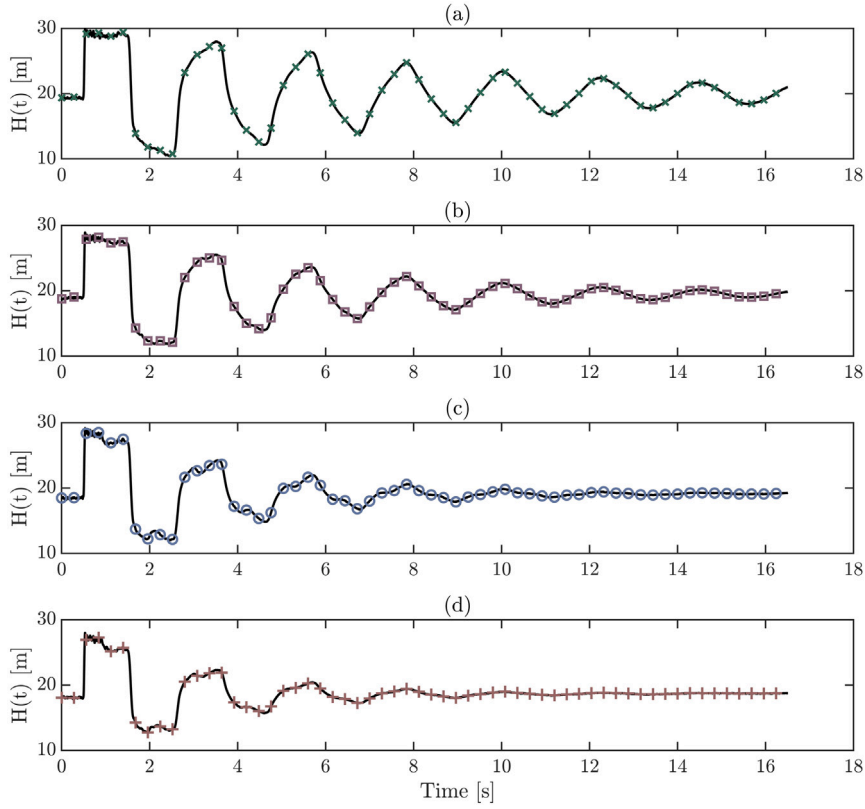


Fig. 14. Pressure head responses in a hydraulic system subjected to valve closure, showing cases of (a) no leak, (b) one leak at x^{L1} , (c) one leak at x^{L2} , and (d) two leaks at both x^{L1} and x^{L2} .

$n = (1, 2, \dots, NP)$ denotes the leak function for n -th pipe branch, the resulting target vector can be organized as

$$\mathcal{L}_T = [\mathcal{L}_1, \mathcal{L}_2, \dots, \mathcal{L}_n, \dots, \mathcal{L}_{NP}]^T$$

where $\top$ denotes the vector transpose, and $\mathcal{L}_T$ is the comprehensive target output vector. Similarly, the input pressure head from various locations can be organized in the following manner: Suppose there are NM sensors, and h_m^j (for $j = (1, 2, \dots, NM)$) represents the pressure head signal from the j th sensor, the resulting input vector can be organized as

$$\mathbf{h}_I = [h_m^1, h_m^2, \dots, h_m^j, \dots, h_m^{NM}]^T$$

where $\mathbf{h}_I$ denotes the comprehensive input vector. Through this approach, the framework can incorporate multiple pipelines and pressure sensors, making the model extendable to a pipeline network. Subsequently, to demonstrate the proposed method's performance, two examples of pipe networks are considered: (1) a system with two pipes connected in series, and (2) a tree-type network with three pipelines, as shown in Fig. 16.

6.2. Example 1: Pipes in series

6.2.1. System description

Consider two pipes connected in series (see Fig. 16(a)), with lengths of 250 m and 300 m, and diameters of 150 mm and 140 mm, respectively. The wall thicknesses of these pipes are 5 mm and 4 mm. Both pipes are elastic, with a Modulus of Elasticity of 170 GPa, resulting in transient wave speeds of 1244 m/s and 1218 m/s in Pipe 1 and 2, respectively. The upstream end of the first pipe is connected to a reservoir, and the downstream end of the second pipe is fitted with a valve, constantly supplied with water at 2 LPS and a pressure of 40 m.

6.2.2. Data generation and NN model development

The sampling frequency is set to 250 Hz, and the total simulation time is defined as $T = 3 \times 4(L_1/a_1 + L_2/a_2)$, where 3 denotes the total number of water hammer wave periods, and 4 characterizes the number of trips the initiated wave makes to complete one cycle.

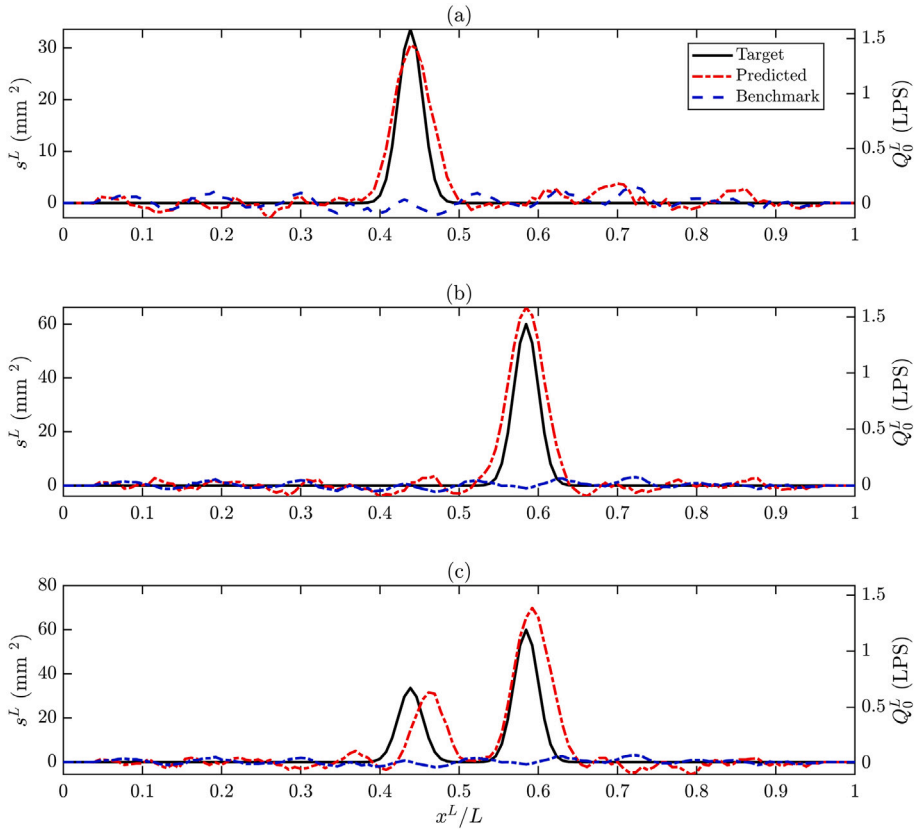


Fig. 15. Comparative plots of true versus predicted leak properties for the ExpModel. Each row corresponds to a different test case: Case 2, Case 3, and Case 4. Leak size s^L is plotted against normalized leak location x^L/L . The solid black line indicates the true values, the dash-dotted red line represents the ANN predictions, and the dashed blue line illustrates the scenario using no-leak data (Case 1) as a referencing threshold.

Datasets are generated under various scenarios: one leak in pipe 1, one leak in pipe 2, one leak in both pipe 1 and pipe 2, and similarly for two and three leaks. As a result, a total of 18,000 (2000 each scenario) samples are simulated and compiled into a dataset. For this paper, the pressure head signal from only one location is utilized, although a higher number of sensor locations can be employed for more complex systems. The simulated pressure signals are not provided here for the sake of brevity.

To develop the model, an ANN model, similar to those used in previous examples, is employed. The model architecture, optimized through a trial and error approach, consists of layers with [400, 400, 300, 200] neurons. All other hyperparameters remain as described in Table 1. The dataset is then pre-processed following the steps described in Section 4.1.1. Given the substantial size of the training dataset, an initial attempt to train the model using the entire dataset simultaneously proved unsuccessful, with training times extending up to 35 h without achieving accurate results. Consequently, the strategy was revised to employ subsets of the data. Accordingly, the dataset was divided into five subsets, each comprising 3600 samples from various scenarios. Adopting this approach, the model was initially trained on subset 1, followed by sequential retraining on the subsequent data subsets. Beginning with subset 2, it was systematically retrained on each successive subset – 3, 4, and finally 5 – with each retraining session spanning 36 to 45 min. A 70:20:10 ratio for training, validation, and testing is maintained throughout. After training on all subsets, the model is applied to the entire dataset (18,000 samples) to ensure that NN weights are adequately adjusted without overlooking earlier subsets. Care is taken to meticulously maintain sample indexing (for training, validation, and testing) during the model development to prevent data leakage. The final training stage required approximately 3 h, culminating in a total training time of around 8 h. Results are presented in the following section.

6.2.3. Results

Fig. 17 presents the comparative results for the pipe-in-series system, with each row representing a different phase – training, validation, and testing – and each column showcasing one of three randomly selected examples that illustrate the model's performance. It can be seen that the ANN model predictions closely align with the true values. The overall efficiency of the model, as quantified by various performance metrics, and the regression plots are showcased in Fig. A.22 (refer to Appendix).

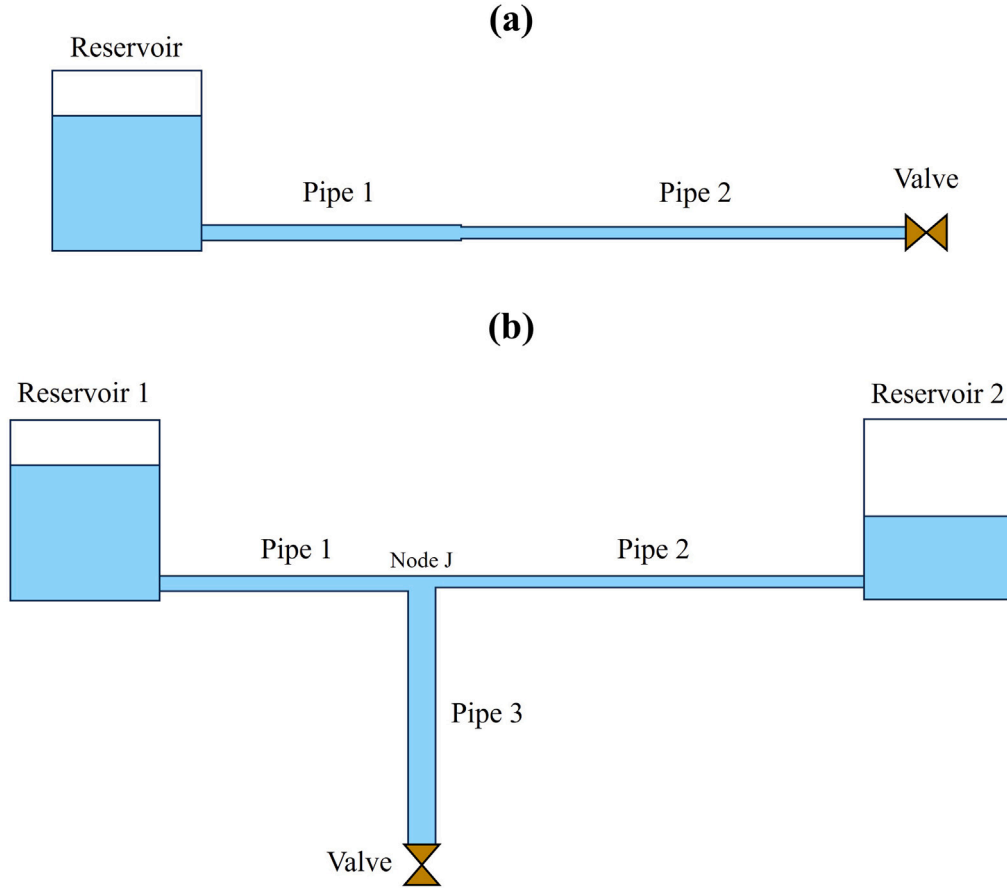


Fig. 16. Schematics of the (a) system with two pipes in series and (b) a tree-type network with three pipelines.

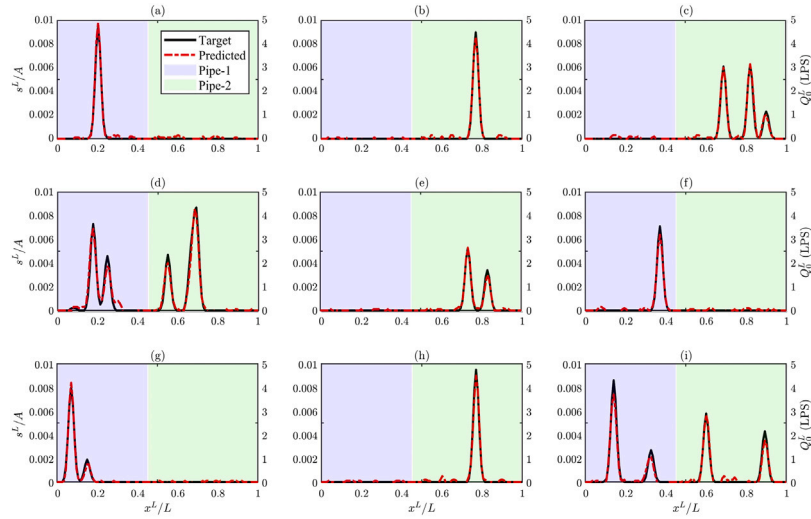


Fig. 17. Comparative plots of true versus predicted leak properties for the pipe-in-series system. Each row corresponds to a different dataset: training, validation, and testing; while each column represents one of the three examples illustrating performance, randomly chosen. The normalized leak size s^L/A is plotted against the normalized leak location $x^L/(L = L_1 + L_2)$. The solid black line indicates the true values, and the dashed red line represents the ANN predictions. The shaded blue region highlights the range of pipe 1, and the shaded green region highlights the range of pipe 2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

6.3. Example 2: Tree-type pipe network

6.3.1. System description

Consider a tree-type pipe network as shown in Fig. 16(b). The lengths of the pipes are $L_1 = 250$ m, $L_2 = 300$ m, and $L_3 = 250$ m, with diameters of 150 mm, 140 mm, and 160 mm, respectively. The wall thicknesses are 5 mm for pipe 1, 4 mm for pipe 2, and 6 mm for pipe 3. All pipes have a modulus of elasticity of 170 GPa, resulting in wave speeds of 1244 m/s for pipe 1, 1218 m/s for pipe 2, and 1234 m/s for pipe 3. The valve at the downstream end of pipe 3 is maintaining a steady-state flow rate of 2 LPS at a pressure head of 40 m. This valve is used for transient generation.

6.3.2. Data generation and NN model development

The sampling frequency is set at 250 Hz. The total simulation time is defined as $T = 3 \times 4(L_1/a_1 + L_2/a_2)$, considering $L_2 > L_3$. Various scenarios are simulated, including a single leak in each of pipes 1, 2, and 3, as well as combinations of leaks across the three pipes, and for two and three leaks simultaneously. In total, 24,000 samples (2000 per scenario) are simulated and compiled into a dataset. The measurement station is located at the downstream end of pipe 3. The simulated pressure signals are not provided here for the sake of brevity.

An optimized ANN model with a structure of [400, 400, 300, 300, 300] is employed. All other hyperparameters are consistent with those described in Table 1. The dataset is pre-processed as outlined in Section 4.1.1. For training, the dataset is divided into four subsets. The model, initially trained on subset 1, undergoes fine-tuning with subset 2, 3, and 4. Following that, the refined model undergoes a final training phase over the entire dataset. The training, validation, and testing split is kept at a 70:20:10 ratio. The entire model development process takes approximately 9.5 h. The subsequent section presents the model's performance and results.

6.3.3. Results

Fig. 18 provides a comparative assessment of the actual versus predicted leak properties by the ANN model for a tree-type pipe system. Each row corresponds to a distinct dataset phase: training at the top, followed by validation, and testing at the bottom. The columns represent three randomly selected examples, illustrating the model's accuracy. It is evident that the ANN maintains high precision in leak localization across all dataset phases. However, there are noticeable discrepancies in leak size estimation, especially for adjacent leaks. Spurious minor lobes are also detected with minimal height in the training phase, and moderate height in validation and testing, possibly indicating the need for denser model with more number of layers/neurons. These findings, in conjunction with the regression analysis shown in Fig. A.22 (refer to Appendix), suggest that the proposed ANN framework is capable and adaptable to complex pipeline systems. The presence of closely spaced peaks indicates that the ANN is sufficiently sensitive to distinguish between proximate leak events, which is essential in dense pipeline networks. Nevertheless, as the complexity of the system increases, there is an evident requirement for an ANN with a higher capacity – potentially more layers and neurons – to effectively model the intricate relationships within the data without compromising the accuracy.

7. Conclusions

A novel domain-guided machine learning-based framework for the localization and sizing of multiple leaks in pressurized pipelines is introduced. A comprehensive numerical dataset, generated using a time-domain hydraulic transient model with the method of characteristics, is utilized to train a feedforward neural network (a variant of the Artificial Neural Network; ANN). This network, featuring a five-layer architecture with 400, 400, 300, 300, and 200 neurons, processes pressure head signals corrupted with Gaussian noise for leak detection. Systematic evaluation of numerical examples demonstrates the ability of the model to accurately predict single and multiple leaks (up to four), amidst modeling uncertainties such as valve time closure, pre-transient pressure head, and flow rate variations. The dataset, consisting of 2000 samples per scenario, is divided into training, validation, and testing subsets in a 70-20-10 split, ensuring the model's robustness and reliability. Experimental validation is performed using data from a Viscoelastic pipe system. The trained ML model, further refined to adapt to various uncertainties, accurately identifies leaks in all test cases, with minor errors in sizing due to modeling errors in the hydraulic transient solver. This validation underscores the model's accuracy and potential for real-world application. The initial training of the baseline model for both numerical and experimental data sets is completed in approximately 2.5 h, with each subsequent refinement requiring an additional hour, indicating the method's efficiency. The computational speed of the model could be enhanced through the use of GPUs.

The proposed framework has also been extended to handle complex pipeline configurations, including systems with two pipes connected in series and tree-type networks with three pipes. This extension involved generating datasets using the hydraulic transient model, optimizing the neural network (NN) model architecture, and strategically training the NN model with small datasets instead of using the entire dataset at once. The framework has shown adaptability to complex systems with moderately accurate results. However, challenges were noted: as system complexity increases, the ANN model's predictions tend to deviate from the expected output. This issue might be addressed by increasing the number of layers and neurons. However, this approach requires careful tuning of the neural network's architecture and the selection of optimal hyperparameters, a process that can be both time-consuming and intricate. Future efforts will focus on exploring more complex pipeline systems and incorporating real-field data, employing diverse or hybrid Machine Learning models to more effectively tackle these challenges.

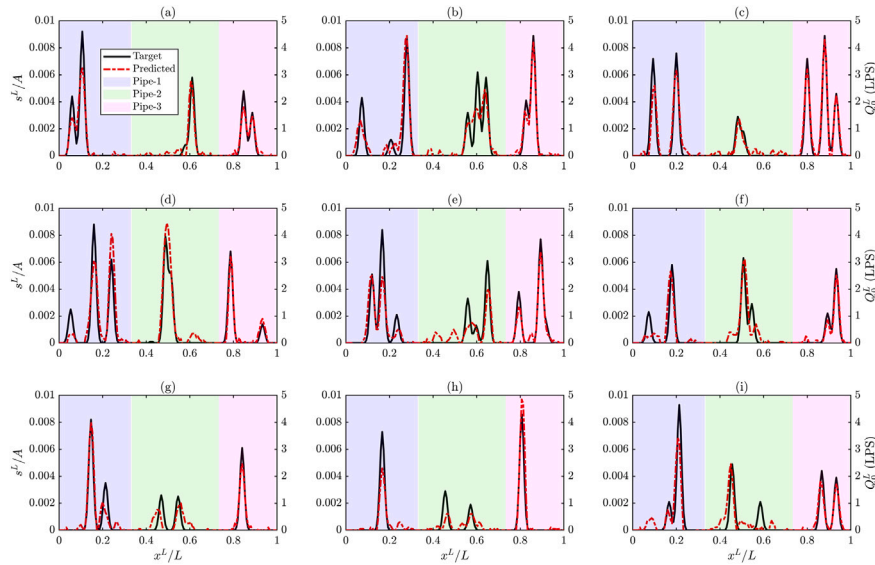


Fig. 18. Comparative plots of true versus predicted leak properties for the tree-type pipe system. Each row corresponds to a different dataset: training, validation, and testing; while each column represents one of the three randomly selected examples illustrating performance. The normalized leak size s^L/A is plotted against the normalized leak location $x^L/(L = L_1 + L_2 + L_3)$. The solid black line indicates the true values, and the dashed red line represents the ANN predictions. The shaded blue, green and red regions highlight the range of pipe 1, 2, and 3, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

CRedit authorship contribution statement

Muhammad Waqar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Azhar M. Memon:** Writing – review & editing, Visualization, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Moez Louati:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Mohamed S. Ghidaoui:** Writing – review & editing, Visualization, Supervision, Methodology, Conceptualization. **Luai M. Alhems:** Software, Resources, Project administration, Funding acquisition. **Silvia Meniconi:** Writing – review & editing, Visualization, Resources, Investigation, Formal analysis, Data curation. **Bruno Brunone:** Writing – review & editing, Visualization, Methodology, Data curation. **Caterina Capponi:** Writing – review & editing, Visualization, Resources, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix. Training performance curves of various ML models

Fig. A.19 showcases the improved predictive performance of AdaptedModel – 1. Notably, the mean squared error (MSE) illustrated in Fig. A.19a does not originate from a high value, indicating that the pre-tuned model parameters only required recalibration. This process was efficiently completed in approximately 5800 epochs and 49 min, meeting the predefined threshold of 1000 validation failures.

The training performance curves of AdaptedModel – 2, comprising the loss function, the loss function gradient, and the validation failures, are graphically summarized in Fig. A.20. This iteration of the model exhibits fine-tuned synaptic parameters, adapting to the combined variability of valve closure and the corresponding Joukowski pressure head in the transient data. Remarkably, this comprehensive tuning was accomplished in just 43 min over 5000 epochs.

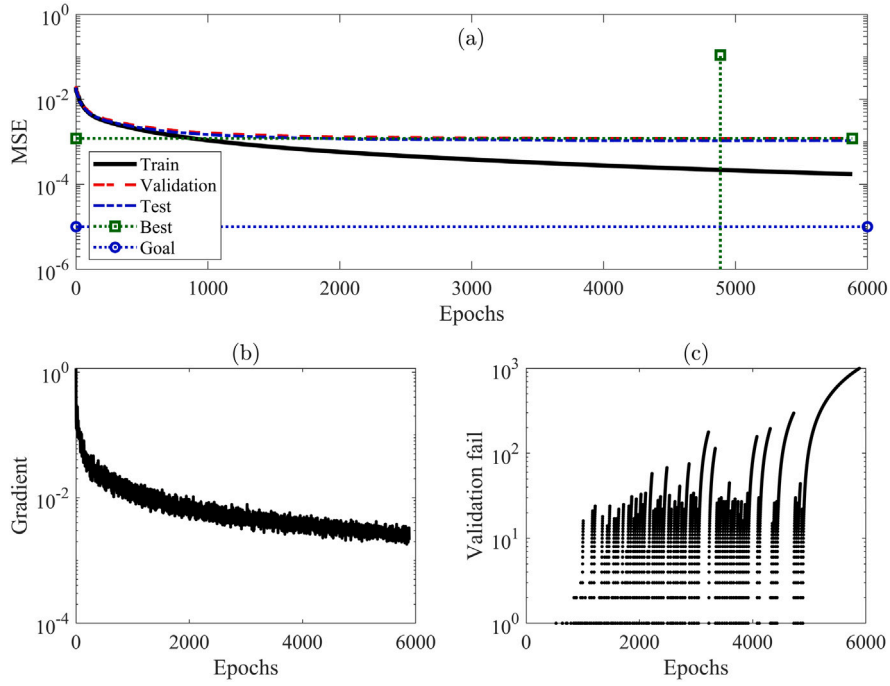


Fig. A.19. Training performance curves for the AdaptedModel - 1. (a) Mean Squared Error (MSE) loss function over epochs for training, validation, and test sets. The 'Best' line indicates the epoch with the lowest validation MSE, and the 'Goal' line represents the targeted MSE threshold. (b) Gradient of the loss function over epochs, demonstrating the optimization process. (c) Validation fails over epochs, with a threshold set to 1000.

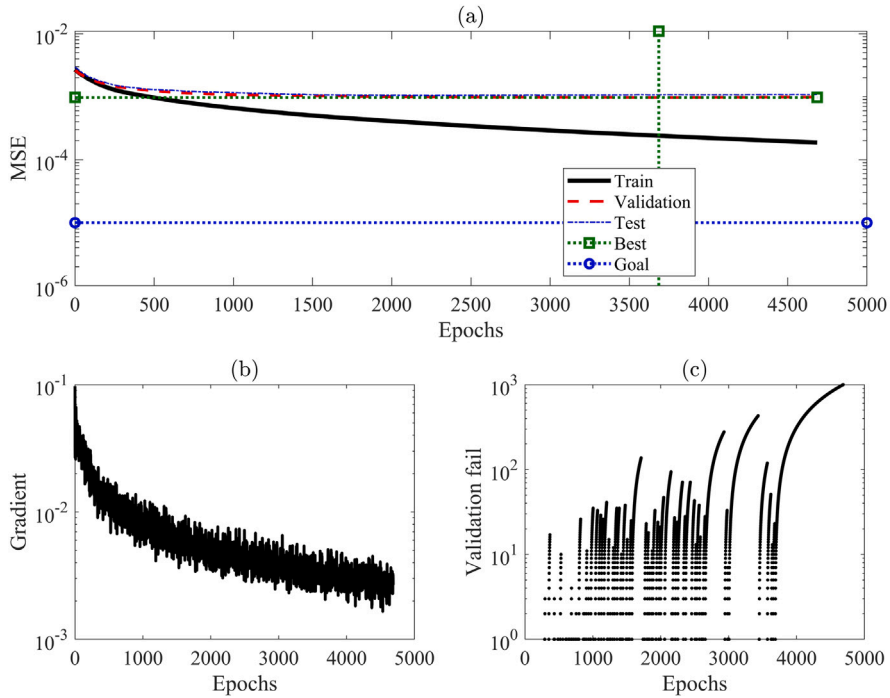


Fig. A.20. Training performance metrics for the AdaptedModel - 2. (a) Mean Squared Error (MSE) loss function over epochs for training, validation, and test sets. The 'Best' line indicates the epoch with the lowest validation MSE, and the 'Goal' line represents the targeted MSE threshold. (b) Gradient of the loss function over epochs, demonstrating the optimization process. (c) Validation fails over epochs, with a threshold set to 1000.

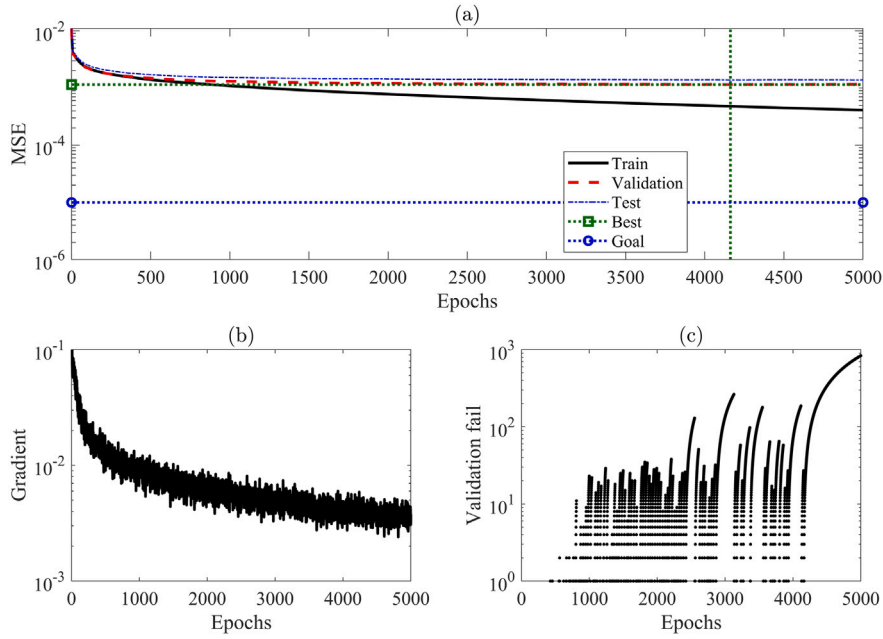


Fig. A.21. Training performance metrics for the AdaptedModel1 – 3. (a) Mean Squared Error (MSE) loss function over epochs for training, validation, and test sets. The 'Best' line indicates the epoch with the lowest validation MSE, and the 'Goal' line represents the targeted MSE threshold. (b) Gradient of the loss function over epochs, demonstrating the optimization process. (c) Validation fails over epochs, with a threshold set to 1000.

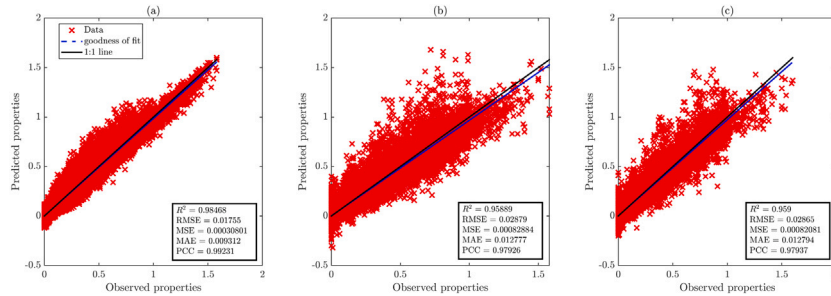


Fig. A.22. Performance metrics for the pipe-in-series system along with regression plots for (a) training, (b) validation, and (c) testing stages. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The training performance of AdaptedModel1 – 3 is depicted through metrics including the loss function, its gradient, and validation failures, as illustrated in Fig. A.21. This version of the model has been calibrated to accommodate the combined variability of valve closure time and pre-transient flowrate, as well as scenarios involving four leaks. Notably, this extensive fine-tuning was achieved within a span of approximately 52 min across 5000 epochs, demonstrating the model's efficiency in adapting to complex hydraulic conditions.

For pipe-in-series- system, Fig. A.22 illustrates the regression analysis of the model across different datasets: (a) training, (b) validation, and (c) testing. Each subfigure plots the predicted properties against the observed properties, with the data points marked in red. The blue line represents the model's goodness of fit, while the solid black line indicates the ideal 1:1 correspondence between the predicted and observed values. Performance metrics such as R^2 , RMSE, MSE, MAE, and PCC are displayed for each dataset, demonstrating the model's accuracy and predictive capability.

For tree-type pipe network, Fig. A.23 presents the regression analysis of the model across different datasets: (a) training, (b) validation, and (c) testing. Each subfigure plots the predicted properties against the observed properties, with the data points marked in red. The blue line represents the model's goodness of fit, while the solid black line indicates the ideal 1:1 correspondence between the predicted and observed values. Performance metrics such as R^2 , RMSE, MSE, MAE, and PCC are displayed for each dataset, demonstrating the model's accuracy and predictive capability.

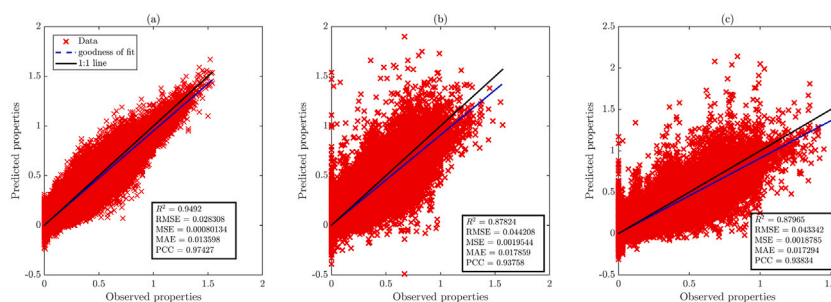


Fig. A.23. Performance metrics for the tree-type pipe system along with regression plots for (a) training, (b) validation, and (c) testing stages. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

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